

Week 4: Activity 2 — Mastering the Box Model & Flexbox

Course Practical Implementation Guide & Source Files

1. Objective & Overview

This activity establishes a core practical understanding of the **CSS Box Model** (comprising Content, Padding, Border, and Margin) alongside **Flexbox Layout Systems**. As outlined in the Weeks 4 & 5 syllabus, mastering these layout rules ensures your structural designs translate gracefully across variable viewports prior to transitioning into full responsive media queries.

You will transform a standard semantic HTML block into a structured component utilizing best-practice architectures (**External Stylesheets**) while maintaining a strict file layout.

Syllabus Requirement Reminder

Always avoid inline style declarations. External files maximize maintainability, isolate layout calculations, and ensure compatibility with modern deployment environments.

2. Mandatory Project Folder Structure

Before writing any code elements, verify your local workspace mirrors the required standard configuration precisely. Deductions apply during Terminal Assessments for missing assets or unorganized flat files.

```
portfolio-activity2/ | ├── index.html └── css/ └── style.css
```

3. Document Source Markup (index.html)

Populate your root layout document with the clean, semantic structural container tags provided below. This establishes the structural blueprint for your component card.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Week 4 Activity 2 - Profile Component</title>
  <!-- Link to external asset -->
  <link rel="stylesheet" href="css/style.css">
</head>
<body>

  <main class="page-wrapper">
```

```
<article class="profile-card">

  <header class="card-header">
    <h1 class="profile-name">Alex Dev</h1>
    <p class="profile-title">Frontend Developer Student</p>
  </header>

  <section class="card-body">
    <p>Learning modern web styling, semantic HTML5, and responsive layouts
during Weeks 4 & 5.</p>
  </section>

  <footer class="card-footer">
    <a href="#" class="btn-link">GitHub</a>
    <a href="#" class="btn-link">Portfolio</a>
    <a href="#" class="btn-link">Contact</a>
  </footer>

</article>
</main>

</body>
</html>
```

4. Cascading Style Sheet Configurations (css/style.css)

The following layout properties isolate the element boxes, enforce specific dimensions, and utilize structural flex wrappers to anchor elements accurately within the display grid.

```
/* Global Initialization Rules */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', sans-serif;
  background-color: #f0f2f5;
  color: #333333;
}

/* Center Container Viewport Alignment */
.page-wrapper {
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
  padding: 20px;
}

/* Structural Application of the CSS Box Model */
.profile-card {
  background-color: #ffffff;
  width: 100%;
  max-width: 400px;
  padding: 30px;
  border: 2px solid #e1e4e8;
  border-radius: 12px;
  margin: 15px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.05);
}

.card-header {
  margin-bottom: 20px;
  text-align: center;
}

.profile-name {
  font-size: 1.8rem;
  color: #1a1a1a;
  margin-bottom: 5px;
}

.profile-title {
  font-size: 1rem;
  color: #0066cc;
  font-weight: 600;
}
```

```

}

.card-body {
  font-size: 0.95rem;
  line-height: 1.6;
  color: #555555;
  margin-bottom: 30px;
  text-align: center;
}

/* One-Dimensional Flex Distribution layout */
.card-footer {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.btn-link {
  text-decoration: none;
  color: #ffffff;
  background-color: #1a1a1a;
  padding: 8px 16px;
  border-radius: 6px;
  font-size: 0.85rem;
  font-weight: 500;
}

.btn-link:hover {
  background-color: #0066cc;
}

```

5. Evaluation Criteria & Submission Instructions

- **Code Validation:** Ensure all HTML structural declarations and tags close securely. Run style code through valid syntax linters to confirm no trailing properties break processing bounds.
- **Repository Setup:** Create a public repository on GitHub named `portfolio-activity2`. Push your code adhering strictly to the required folder pattern.
- **Documentation Requirements:** Capture high-resolution window screenshots mapping your complete application runtime inside a browser viewport alongside code views within your editor. Package your code screenshots alongside your active repository hyperlink into a unified documentation file.